CLAUDE CERTIFIED ARCHITECT — FOUNDATIONS

Objective 25: Structured Output Method

Exam objective: Select and implement the most reliable structured output method—tool use with JSON schema, prompt-based formatting, or prefilled responses—based on required schema compliance strictness.

Q1 **You are building a workflow where model output must always populate a downstream system expecting exact JSON keys, types, and required fields. Which method is most reliable?**

A Prompt-based formatting alone

B **Tool use with a JSON schema**

C Free-form natural language with examples

D Prefilled prose paragraphs

CORRECT ANSWER **B**

WHY When strict schema compliance is required, tool use with a JSON schema provides the strongest structural guarantee. Prompt-only formatting is less reliable because the model can drift from exact syntax or required fields.

Q2 A prototype internal dashboard needs lightweight structured output, but occasional formatting cleanup in application code is acceptable. Which method is usually sufficient?

A Tool use with schema is always mandatory

B Prompt-based formatting can be appropriate when strict compliance is not critical

C Prefilled responses are the only option

D Avoid structure entirely

CORRECT ANSWER **B**

WHY Prompt-based formatting is often acceptable for lower-stakes scenarios where some post-processing tolerance exists. It is less strict than schema-backed tool use but simpler for lightweight needs.

Q3 You want the model to answer with one of three exact labels: APPROVE, REJECT, or ESCALATE. Which method is often a good fit?

A Prefilled response starting the answer in the required form

B A long essay prompt

C Random sampling of outputs

D No constraints because the labels are obvious

CORRECT ANSWER **A**

WHY Prefilled responses are useful when the desired output shape is very constrained and small, such as fixed labels or a known prefix. A full JSON schema would also work, but prefill is a simple strong pattern for tightly bounded outputs.

Q4 **Which scenario most strongly argues against relying only on prompt-based formatting?**

A The output is displayed directly to a human who can interpret minor variations

B Downstream automation breaks if one required JSON field is missing or a type is wrong

C The response is short

D The user prefers concise answers

CORRECT ANSWER **B**

WHY If a single structural deviation can break downstream automation, you need stronger guarantees than prompt wording alone. Schema-backed tool use is the safer design in that case.

Q5 **A model must extract invoice data into a validated object with fields for vendor, invoice_number, due_date, and total_amount. Some fields are required and must have the correct types. What is the best implementation choice?**

A Ask for a markdown table

B Use tool use with a JSON schema defining required properties and types

C Ask the model to "be careful with formatting"

D Return plain English and parse it later with regexes only

CORRECT ANSWER **B**

WHY Required properties and type constraints point directly to schema-backed tool use. Tables and free-form text are weaker contracts and increase downstream parsing risk.

Q6 **You need a yes/no classifier result embedded inside a short templated sentence sent to a human reviewer. Which method may be most practical?**

A Tool use with a deeply nested JSON schema

B Prefilled response or prompt-shaped formatting to constrain the short template

C No structure, just a paragraph

D Message Batches API only

CORRECT ANSWER B

WHY For a tiny, human-facing response with limited variants, prefill or simple prompt formatting can be practical and sufficient. A full schema may be unnecessarily heavy if machine validation is not the primary need.

Q7 **An engineer says, "Prompt-based formatting is enough for all structured outputs." What is the best rebuttal?**

A Correct; schemas are never needed

B It depends on strictness requirements—prompting may be fine for soft structure, but strict machine-consumable outputs usually need schema-backed tool use

C Prompting only works for long responses

D Structured outputs are impossible with model APIs

CORRECT ANSWER B

WHY The right method depends on how costly deviations are. Soft structure can tolerate prompt-based formatting, while strict integrations generally require stronger enforcement mechanisms.

Q8 Which requirement is the clearest signal to choose tool use with JSON schema?

A The output must be entertaining

B The output will be read once by a human and discarded

C The output must conform to an exact schema for automated downstream processing

D The prompt is under 100 tokens

CORRECT ANSWER C

WHY Exact machine-consumable schema conformance is the strongest reason to use schema-backed tool calls. Human-only consumption and stylistic preferences do not create the same strictness requirement.

Q9 You need the model to emit a two-field object most of the time, but occasional malformed JSON is acceptable because a retry layer can recover. Which method might be acceptable if simplicity is prioritized?

A Prompt-based formatting

B Only natural-language paragraphs

C A screenshot-based workflow

D Eliminate validation entirely

CORRECT ANSWER A

WHY If the application can tolerate occasional reformatting failures and has recovery logic, prompt-based formatting may be an acceptable trade-off. The key is that strict compliance is not mandatory on every first attempt.

Q10 What is the main advantage of prefilled responses compared with fully unconstrained prompting?

- **A** **They nudge the model into a desired output shape by starting the structure for it**
- **B** They validate types automatically
- **C** They eliminate the need for application logic
- **D** They always outperform schemas

CORRECT ANSWER **A**

WHY Prefill works by biasing completion toward a known structure or label space. It is a shaping technique, not a full validation mechanism like a schema.

Q11 A regulator requires every generated decision object to include `decision`, `justification_code`, and `review_required` with exact boolean and enum constraints. Which method best satisfies the requirement?

- **A** Prompt-based formatting with an example
- **B** **Tool use with JSON schema enforcing required fields and allowed values**
- **C** Prefilled natural language sentence
- **D** Any method because the model usually follows instructions

CORRECT ANSWER **B**

WHY Regulatory constraints and exact field/value requirements call for the most reliable structured-output mechanism available. Schema-backed tool use provides explicit enforcement of required fields and allowed types or enums.

Q12 Which use case is the best match for prefilled responses rather than a full schema?

A Returning a complex nested purchase-order object for direct database insertion

B Producing one of several fixed triage labels at the start of the answer

C Emitting a validated numeric array with range constraints

D Writing a machine-checked audit record with required nested objects

CORRECT ANSWER B

WHY Prefill excels when the output space is small and shape-constrained, such as fixed labels or templated short responses. Complex validated objects are better handled with schema-backed tool use.

Q13 Why is prompt-based formatting considered less reliable than schema-backed tool use for strict outputs?

A It prevents the model from producing JSON at all

B The model may omit fields, change key names, or introduce formatting drift despite instructions

C It only works in batch mode

D It requires too many tokens to ever be practical

CORRECT ANSWER B

WHY Prompt-only structure relies on instruction following rather than explicit structural enforcement. Even good models can occasionally drift in key names, punctuation, or required fields under prompt-only control.

Q14 You are designing an extraction system with three tiers: strict production integrations, semi-structured analyst tools, and fixed-label approvals. Which mapping is best?

A Prompt formatting for strict integrations, free text for analyst tools, and schemas for fixed labels

B Schema-backed tool use for strict integrations, prompt formatting for semi-structured analyst tools, and prefilled responses for fixed-label approvals

C Prefill for everything

D Schema-backed tool use only for human-readable summaries

CORRECT ANSWER B

WHY This mapping aligns method strength to output strictness. Strict integrations need schemas, analyst tools often tolerate softer structure, and fixed-label tasks are well served by prefill patterns.

Q15 Which statement best summarizes how to choose among tool use with schema, prompt formatting, and prefilled responses?

A Always use the simplest method regardless of failure cost

B Choose based on required schema compliance strictness and how costly output deviation is for the downstream workflow

C Use prefilled responses only for long essays

D Structured output choices are unrelated to system design

CORRECT ANSWER B

WHY Output-method selection should be driven by strictness requirements and the operational cost of malformed output. Stronger guarantees justify stronger mechanisms, while lighter-weight methods are fine when deviation is tolerable. --